Research Question: predicting diabetic patients
Data Source: NHANES data (2009-2012 cycles)
Goal Metrics:

- Primary: recall (tp/tp+fn)
    - Selected because the cost of misdiagnosis is high
    - We don’t want to miss patients
- Supporting: precision(tp/predicted positive) + f1 +prauc

Takeaway

- Evaluation, your understanding–threshold, intuition is to change xxx
- Test threshold 0.5 versus model learning of optimal threshold
- Deal with imbalance dataset, large missing data
- L1 versus L2, is that L2 will drop features; tree non-parametric and non-linearity

Data:

- Interest variable – whether or not patient has diabete
    - Recoded into 0 (no diabete),1(with diabete)--binary classification task
- Feature variable –demographic, physiological information
    - Demographic: gender, age, homeown, income, etc
    - Physiological: weight, height, BMI, BPSysAve, UrineFlow1, etc
    - Some variables are gender-specific: Age1stBaby, nPregnancies, nBabies
- Imbalance by how much

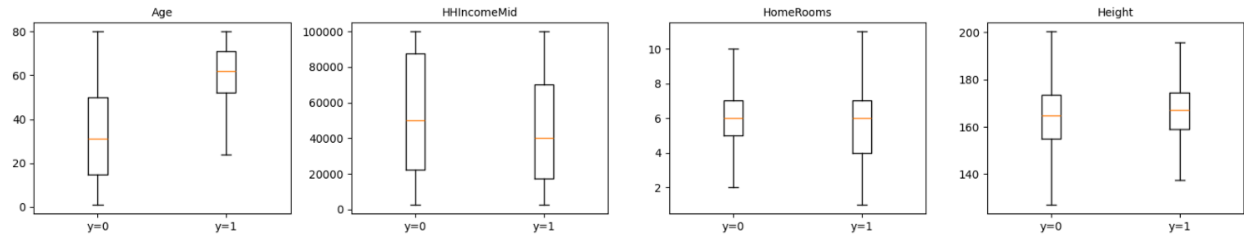
Preprocessing

- Basic data cleaning, keeping only 1 ID with the earliest survey year
    - Why—-across year could have different, might use year as an indicator; don’t want panel data, only cross-sectional
- Check for non-zero variance
- EDA
    - Missing % + Skewness + Correlation with target + Multicollinearity

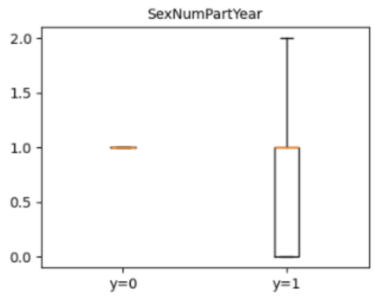
```
VARIABLE SUMMARY (Missing % + Skewness + Corr w/ Target)
================================================================================
       variable  missing_%  correlation  abs_corr  abs_skewness
            Age       0.00     0.317106  0.317106      0.251045
            BMI       2.71     0.238033  0.238033      0.963458
         Weight       0.83     0.208802  0.208802      0.058568
       BPSysAve      15.23     0.205626  0.205626      1.052141
DaysPhysHlthBad      26.51     0.176562  0.176562      2.604528
        nBabies      76.31     0.158825  0.158825      1.602923
   nPregnancies      74.52     0.149085  0.149085      3.164334
==========================================================
HIGH MULTICOLLINEARITY (|r| > 0.8)
==========================================================
       var1    var2  abs_corr_pair drop_suggestion
     Length  Height          0.988          Length
     Weight     BMI          0.903          Weight
HHIncomeMid Poverty          0.903         Poverty
        Age  Length          0.850          Length
     Weight  Length          0.826          Length
```

    - Removed collinear variables + high missing values due to unmeasurement

- Boxplot for visualizing skewness and correlation



- Fixing data:
  - Feature engineering:
    - Removed SexNumPartYear



    - Log transformation on variables with skewness
    - Gender specific variables recoded
    - One-hot encoding for categorical variables
  - Missing data handling (Experiment 1: Complete-case analysis)
    - Low recall, only 3 out of 5 folds were calculated
    - Result show that imputation is needed

Model	Folds	Recall	Precision	F1	PRAUC	Thr	Recall_SD	Precision_SD	F1_SD	PRA
Logistic	3	0.555556	0.132275	0.200000	0.352116	0.157890	0.41574	0.117598	0.163299	0.
XGBoost	3	0.555556	0.155844	0.235897	0.396970	0.039751	0.41574	0.118079	0.171009	0.

  - Imputation + class imbalance handling(Experiment 2)
    - Baseline model performance: 0.077
    - Tree models should use oversampling
    - Logistic should use no class imbalance handling but might differ if combined with L1/L2

Model	Imbalance	Folds	Recall	Precision	F1	PRAUC	Thr	Recall_SD	Precisio
XGBoost	oversample	5	0.875152	0.216362	0.345490	0.382550	0.169236	0.046464	0.02
XGBoost	scale_pos_weight	5	0.867846	0.222299	0.353600	0.377863	0.176176	0.026088	0.01
XGBoost	none	5	0.857068	0.231474	0.362558	0.398906	0.050593	0.065833	0.02
Logistic	none	5	0.837297	0.236565	0.366588	0.387650	0.072617	0.083376	0.02
Logistic	oversample	5	0.831826	0.233781	0.359994	0.376099	0.427947	0.100429	0.03
XGBoost	smote	5	0.824472	0.261920	0.395522	0.411513	0.084920	0.066822	0.02
Logistic	smotenc	5	0.822604	0.178114	0.291417	0.308605	0.178051	0.063131	0.02
Logistic	class_weight	5	0.820917	0.251686	0.383602	0.385554	0.482815	0.079221	0.01
Dummy_stratified	n/a	5	0.077838	0.075439	0.076619	0.082839	0.500000	0.021670	0.02
Dummy_most_frequent	n/a	5	0.000000	0.000000	0.000000	0.082970	0.500000	0.000000	0.00

Model Choice(Experiment 3)

- Models examined:
  - Baseline model
  - Tree model + oversampling: XGB, AdaBoost, RF
  - Logistic models + different class imbalance handling: Logistic, +L1,+L2

- Results:
    - Tree: XBG +oversampling had highest recall, 0.875
    - Logistic: Logistic_L1 +oversampling had highest recall, 0.846

Model	Imbalance	Folds	Recall	Precision	F1	PRAUC	Thr	Recall_SD	Precision_SD
XGBoost_none	oversample	5	0.875152	0.216362	0.345490	0.382550	0.169236	0.046464	0.023806
XGBoost_l1	oversample	5	0.871597	0.225174	0.355963	0.384560	0.187349	0.061059	0.026002
AdaBoost	oversample	5	0.853481	0.234817	0.367258	0.382885	0.488045	0.059563	0.016839
XGBoost_elastic	oversample	5	0.851679	0.236040	0.367278	0.398662	0.230721	0.071021	0.026731
XGBoost_l2	oversample	5	0.847961	0.239901	0.372246	0.390379	0.245675	0.061837	0.023783
Logistic_l1	class_weight	5	0.846388	0.232184	0.361449	0.385987	0.427711	0.096574	0.025488
RandomForest	oversample	5	0.840753	0.254275	0.389295	0.378225	0.147333	0.043873	0.027827
Logistic_logistic	none	5	0.837297	0.236565	0.366588	0.387650	0.072617	0.083376	0.023376
Logistic_l2	none	5	0.837297	0.236565	0.366588	0.387650	0.072617	0.083376	0.023376
Logistic_elastic	none	5	0.835495	0.239720	0.370143	0.389563	0.075860	0.084601	0.023689
Logistic_elastic	class_weight	5	0.835479	0.246346	0.378474	0.385507	0.459388	0.088166	0.021814

- Hyperparameter tuning (Experiment 4)
    - Model choice: XBG +oversampling
    - Nested CV instead of validation set splitting
    - Results showed lower recall but under a different threshold
    - Threshold high, need more evidence to say the person is diabetic

Model	OuterFolds	Recall	Precision	F1	PRAUC	Thr	Recall_SD	Precision_SD
XGBoost_oversample_tuned_nestedCV	5	0.860557	0.229479	0.36196	0.395607	0.205731	0.013054	0.02053

- Parameter refitting (Experiment 5)
    - Threshold: 0.205731
    - Exp 4: chosen from fold 4, highest recall &F1
      {'colsample_bytree': 0.8, 'gamma': 0.0, 'learning_rate': 0.03, 'max_depth': 4, 'min_child_weight': 1, 'n_estimators': 300, 'reg_alpha': 0.0, 'reg_lambda': 5.0, 'subsample': 1.0}
    - Exp 3:
      n_estimators=300, max_depth=4, learning_rate=0.05, subsample=0.9, colsample_bytree=0.9, eval_metric="logloss", n_jobs=-1, random_state=RS, reg_lambda=1.0, reg_alpha=0.0
    - Results:
        - Exp 4 parameter performed better.

=== CV results (fixed THR) ===

Model	Folds	Recall	Precision	F1	PRAUC	Thr	Recall_SD	PRAUC_SD
XGB_param1	5	0.902326	0.200719	0.328258	0.401424	0.205731	0.039859	0.034500
XGB_param2	5	0.849762	0.229560	0.361326	0.382550	0.205731	0.028849	0.033975

- Deployment
    - Test the tuned data